

Unit 1 Choosing a Topic

1 Critical Reading

Understanding the text

TASK 1

Key words	Laypeople, radiation, understanding
Objective	To investigate lay people's knowledge of radiation phenomena and risk
Methods used	Questionnaires and interviews

TASK 2

- 1 Respondents were aware of the harmful health effects of radiation, while they had a little understanding of the concept of radiation.
- 2 There are a big proportion of respondents who support to continue the production of nuclear power.
- 3 The mass media plays a very important role in the misunderstanding of radiation.
- 4 A school program should be designed to analyzing everyday conceptions about radiation.

TASK 3

Paras. 1-2	e	f	b	a	c	h	g	d		
Paras. 3-6	k	h	c	d	a	b	e	j	f	i g
Para. 7	b	c	h	a	d	f	e	g		
Paras. 8-11	f	a	i	k	h	g	c	j	b	d e
Paras. 12-18	d	h	g	f	a	b	c	e		
Paras. 19-20	g	c	a	b	i	d	h	e	f	
Paras. 21-28	i	j	a	h	k	b	c	g	d	e f

TASK 4

1-5	2	5	4	1	3
6-10	9	7	10	6	8
11-15	13	15	11	14	12
16-20	20	19	18	17	16
21-25	25	23	21	24	22
26-28	27	26	28		

TASK 5

- 1 The answer reveals that the respondents fail to understand that a new nucleus will be produced when a radioactive atom is broken.
- 2 According to the view of constructivists, a person's ideas will survive school education and form an important basis for new learning.
- 3 The solution is to consider lay ideas and combine factual knowledge with practical or political factors.

TASK 6

Different types of radioactive elements will make different health effects, according to characteristics of each type.

TASK 7

It may be argued that the population should have a sufficient understanding of radiation phenomena to secure individual safety as well as democratic decisions. To attain this, it is necessary for the communicators of radiation information to be familiar with the laypeople's perceptions of these phenomena. A questionnaire survey was conducted to examine non-experts' conceptions of radiation phenomena. The survey revealed an incomplete understanding of concepts such as **radioactive decay**, **half-life** and **absorption of radiation** and a lack of differentiation of between radiation and radioactive materials. Many respondents did not distinguish between sources of ionising radiation and other environmental hazards. Nuclear power plants and submarines were the most feared sources of radiation. There are indications that the lay understanding of radiation phenomena and risk is to a large extent formed by mass media and that "school knowledge" of these phenomena is not applied in situations belonging to the "real world". To resolve this problem, new teaching and information procedures are needed. These should take into account the learners' perceptions and should integrate "school knowledge" with considerations belonging to the "real world".

Enhancing language ability

TASK 2

- 1 betrays, conception
- 2 renders
- 3 exemplified, perception
- 4 prevalent, secure
- 5 prior to
- 6 proportion, gloomy
- 7 alternatives
- 8 characterized, constitutes, hazard
- 9 From the standpoint of, ranks
- 10 external, a prerequisite

TASK 3

- 1 **include:** contain, embrace, encompass, comprise, be composed of, consist of
- 2 **valid:** powerful, convincing, sensible, rational, viable, credible, plausible, weighty, well-founded, cogent, well-grounded
- 3 **basic:** underlying, fundamental, essential, root, prime, primary
- 4 **danger:** hazard, risk, threat, menace, peril, jeopardize
- 5 **describe:** depict, portray, characterize, narrate, outline, represent
- 6 **different:** discrepant, contrary, conflicting, contradictory, inconsistent, incompatible, incongruous, various, diverse
- 7 **cause:** prompt, move, inspire, stimulate, urge, spur, motivate, induce, evoke, instigate, impel
- 8 **clarify:** account for, justify, explain, illuminate, clear up, rationalize, elucidate, explicate

- 9 **highlight:** emphasize, stress, accent, feature, underline, spotlight, accentuate, give prominence to, attach importance to, give priority to, underscore
- 10 **factor:** component, item, element, ingredient, constituent, contributor

Doing research projects

TASK 1

- 1 Solvent, pesticide, smoke, viral aggressions, ultrave radiation and ionizing radiation.
- 2 Cells could identify the DNA damage, and remove the damaged part and re-synthesis to repair the DNA.
- 3 The Stochastic effects are not systematic to each person being exposed. Their development could not be predicted by the scale of individual which is opposite from the determinist effect.
- 4 The healthy surroundings may be exposed to radiation during the treatment.
- 5 Radiation dose should be keep as low as possible.

2 Academic Writing

Understanding a research report or research article

TASK 2

- 1 Four. They are problem section, procedures section, result section and discussion section.
- 2 Because it will contribute to the credibility of the research.
- 3 It can allow the reader to duplicate the experiment if there is some doubt about your findings.
- 4 It may evaluate the research results fully, point out what questions remain unanswered and perhaps suggest directions for further research.
- 5 Accuracy, clarity, and completeness.

Deciding on a topic

TASK 2

- 1 **Genetically Engineering** → The Impact of Genetically Engineering → The Impact of Genetically Engineering on Agricultural Practice / Food Shortage
- 2 **Breast-feeding Practice** → Breast-feeding Practice in Chinese Big Cities → The Impact of Urbanization on Breast-feeding Practice in Chinese Big Cities
- 3 **Internet Technology** → The General Impact of Internet Technology on Education → The Positive Impacts of Internet Technology on Teaching Methodology

TASK 3

- 1
- 1) Vulnerabilities and the Risk from Global Warming
- 2) Social Issues and Global Warming
- 3) The Impacts of Global Warming on Food Products

2

- 1) Genetically Modified Foods and Health Issues
- 2) Controversies as to Genetically Modified Foods
- 3) The Development of Genetically Modified Foods in China

3

- 1) The Future of Nuclear Energy
- 2) Safety Issues about Nuclear Power Plants
- 3) Waste Management and Nuclear Power Plants

4

- 1) Science Fraud in the Medical Field
- 2) Science Fraud in China
- 3) Research Pressure and Science Fraud

Formulating research questions

TASK 1

1 Nuclear Waste

- 1) What is nuclear waste?
- 2) Why is it difficult to dispose of nuclear waste?
- 3) What harmful effects if nuclear waste is not appropriately disposed of?
- 4) How to dispose of nuclear waste safely and economically?

2 Threats of Artificial Intelligence

- 1) What are the present threats of artificial intelligence?
- 2) What are the future threats?
- 3) How can human beings control those threats?

3 Global Warming and Its Effects

- 1) What is Global warming?
- 2) How does it affect agriculture?
- 3) How does it affect our environment?
- 4) How does it affect our health?

4 Genetically Modified Foods

- 1) What are advantages and disadvantages of GMFs?
- 2) Why do GMFs gain more popularity in the developing countries?
- 3) How safe are GMFs?
- 4) What are the effects of GMFs on human health?

TASK 2

My topic: Science Fraud Epidemic in China		
My Reasons	1	It is prevalent in universities and scientific institutes.
	2	There are many reports and articles discussing science fraud.
	3	It is manageable since the study involves questionnaire and interviews.

Research Questions	1	What is the current situation of science fraud in tertiary institutions?
	2	What are the underlying causes for the prevalence of science fraud?
	3	What measures could be taken to address the problem?

Writing a working title

TASK 2

- 1 Question titles are not appropriate. It is better to use "Economic Effects of Climate Change"
- 2 Good
- 3 Impacts of Global Warming on China
- 4 Good
- 5 Good
- 6 Good
- 7 Good

3 Literacy Skills

Avoiding plagiarism

TASK 1

- 1 Y
- 2 Y
- 3 Y
- 4 Y
- 5 Y
- 6 Y
- 7 N
- 8 Y
- 9 Y
- 10 Y

TASK 3

- 1 Not common knowledge
- 2 Not
- 3 Not
- 4 Not
- 5 Not

Citation

TASK

Numerous research studies demonstrate consistently that content-based second language teaching promotes both language acquisition and academic success (Grabe & Stoller, 1997; Kasper, 1994; Krueger & Ryan, 1993; Snow & Brinton, 1997; Stryker & Leaver, 1997; Wesche, 1993). Students receiving linked instruction perform better in language courses than those not receiving such instruction (Kasper, 1997). They reap the benefits of significant

gains in the second language, e.g., in the receptive skills of listening and reading ([Burger et al., 1997](#); [Ready & Wesche, 1992](#)) and in the productive skills of writing ([Burger, 1989](#)) and speaking ([Burger & Chrétien, 2001](#)). They also achieve comparable or even better mastery of disciplinary content than ESL students or native English-speaking students not receiving content-based language instruction ([Andrade & Makaafi, 2001](#); [Babbitt, 2001](#); [Kasper, 1994](#); [Winter, 2004](#)).

The topic sentence is “content-based second language teaching promotes both language acquisition and academic success.” The writer uses literature evidence to support the idea in terms of speaking, reading and writing.

Quoting

TASK 1

- 1 **Author-focused version:** According to Anyanwu (1983), most Nigerians believe that “urban existence is synonymous with extreme individual and community poverty”.
Information-focused version: Influenced by western cultures, most Nigerians believe that “urban existence is synonymous with extreme individual and community poverty” (Anyanwu, 1983).
- 2 **Author-focused version:** Anyanwu (1983) suggested that mothers should be educated “how to prepare and use hygienically acceptable supplementary foods”.
Information-focused version: We may educate mother “how to prepare and use hygienically acceptable supplementary foods” (Anyanwu, 1983).

Summarizing

TASK 1

- 1 c. b. a. d
- 2 b. a. c. d
- 3 a. d. c. b

TASK 2

- 1 Chinese big cities suffer from growing traffic problems.
- 2 Some long and complicated sentences mean very little.
- 3 The basic tenet of constructivism holds that learners themselves construct knowledge, rather than receive it from outside.
- 4 Some educationalists contend that deciding on a major is the most critical decision students face when entering college, whereas others argue that the most important thing for students to discover in university is their true interests, because many establish careers not related to the major they studied in college.
- 5 The majority of American Internet users, especially teenagers, use SNS, which has altered the pace and process of making and maintaining friendships.
- 6 According to Hirshi and Gottfredson, the propensity (倾向) of someone to commit crime comes from his or her lack of self-control, whereby the urge to feed desires causes one to even break the law.

TASK 3

- 1** Gordon & Taylor (1989) believes that although most people feel a great deal of anguish when faced with writing tasks, this feeling can be managed by developing personal procrastination-avoidance strategy.
- 2** Frick (1991) claims that history has demonstrated that technology affects education profoundly. Considering the definition of technology broadly, one may say that prehistoric people used primitive technologies to teach skills to their young.
- 3** Hewitt (1996) believes that acts of academic dishonesty undermine the validity of teaching and learning.
- 4** Edelson (2002) claims that legal rights and safeguards lead to investments by individuals, which has a higher impact on raising living standards of countries than these countries' natural resources have.



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Unit 2 Searching for Sources

1 Critical Reading

Understanding the text

TASK 1

Key words	Interactive Technology, Students' Understanding, Greenhouse Effect, Global Warming
Objective	to report how students' understanding of the greenhouse effect and global warming has been increased after they participated in an experiment based on a curriculum unit
Method used	experiment

TASK 2

- Students conducted virtual experiments with a visualization of the greenhouse effect. They manipulated the experiment, analyzed data and drew conclusions about how individual variables affect changes in the Earth's temperature. They also carried out inquiry activities to make connections between scientific processes. The procedure section tells it.
- The goal is for students to learn about the role of each concept in the greenhouse effect and to understand how they interact to create the greenhouse effect
- The function of the two tests is to measure the changes in students' understanding the greenhouse effects. The content of the two tests is the same: there are eight questions testing the participants' knowledge of the greenhouse effects. This can be informed in the research design section.
- To compare their results with theirs and find a similar conclusion that children tend to reconcile new ideas with their pre-existing knowledge.

Task 3

Paras. 1-2

- ~~exploration~~
- ~~insert~~
- ~~misleading~~
- ~~contribute~~
- ~~completion~~
- ~~visual~~
- ~~syllabus~~
- ~~sector~~

Paras. 3-6

- ☐
- ~~chart~~
- ~~impose~~
- ~~attack~~
- ☐
- ~~mirror~~
- ☐

Paras. 7-9

- 1 ~~irregularly~~
- 2 ☐
- 3 ~~repeat~~
- 4 ~~promote~~
- 5 ~~numerous~~
- 6 ☐
- 7 ☐
- 8 ☐

Paras. 10-20

- 1 ☐
- 2 ~~contribution~~
- 3 ☐
- 4 ☐
- 5 ☐
- 6 ☐
- 7 ~~normal~~
- 8 ☐

Paras. 21-25

- 1 ☐
- 2 ~~honesty~~
- 3 ~~join~~
- 4 ~~enlarging~~
- 5 ~~immediate~~
- 6 ~~agree with~~
- 7 ~~conditional~~

TASK 4

- 1-5 5 2 3 4 1
- 6-10 8 10 6 7 9
- 11-15 13 14 15 12 11
- 16-20 19 16 20 18 17
- 21-25 21 23 25 24 22

TASK 5

- 1 According to Meadows and Wiesenmayer's study, some students' misunderstanding of the greenhouse effect is becoming part of their reasoning of global warming.
- 2 Knowledge has accumulated gradually as their understanding of the ideas and connection between ideas has increased.
- 3 Vosniadou and Brewer find that children are characterized by a mental transition from one concept to another when they begin to doubt the earth is flat and accept that it is a sphere.

TASK 6

Although carbon dioxide largely harms the environment, the other greenhouse gases have a large impact on the atmosphere.

TASK 7

In this work, we examined middle school students' understanding of the greenhouse effect and global warming. We designed and refined a technology-enhanced curriculum module called Global Warming: Virtual Earth. In the module activities, students conducted virtual experiments with a visualization of the greenhouse effect. They analyzed data and drew conclusions about how individual variables effect changes in the Earth's temperature. They also carried out inquiry activities to make connections between scientific processes, the socio-scientific issues, and ideas presented in the media. Results show that participating in the unit increases students' understanding of the science. We discuss how students integrate their ideas about global climate change as a result of using virtual experiments that allow them to explore meaningful complexities of the climate system.

Enhancing language ability

TASK 2

- 1 underlying, embedded
- 2 being modified, incorporated
- 3 manipulated, reconcile
- 4 incremental / gradual, outcome, assess
- 5 depicts
- 6 discrepancy, contributor
- 7 attributed, target
- 8 highlights, misconception, impact
- 9 Fluctuation, intermediate
- 10 Approximately, integrated / incorporated

TASK 3

- | | | | | |
|---------|--------------|-----|-----|------|
| 1 event | 2 administer | 3 ○ | 4 ○ | 5 ○ |
| 6 ○ | 7 ○ | 8 ○ | 9 ○ | 10 ○ |

Doing research projects

TASK 1

- 1 It refers to the greenhouse gases released into the atmosphere because of an individual's daily activities such as burning fossil fuels for transportation or heating homes.
- 2 Carbon dioxide CO₂, methane CH₄, nitrous-oxide N₂O.
- 3 It is the measurement for describing how much global warming each greenhouse gas may contribute to.
- 4 A term used to describe the potential greenhouse effect of the agriculture product.
- 5 Car pooling (get more people in one car), using the most efficient route, reducing the idling time, maintaining the vehicle and checking the tire pressure at least once a month, consideration of buying smaller, eco or hybrid version, electric cars, traveling by trains.

2 Academic Writing

Using reading strategies for specific information

TASK

- 1 **Objective:** To evaluate students' knowledge of the Greenhouse Effect and its related issues
Research questions: What is university students' understanding of the Greenhouse Effect and its related issues? Why is their understanding desirable or undesirable?
Methods: Questionnaire and interviews
- 2 **Objective:** To investigate elementary school children's explanations of the day / night cycle
Research questions: What are explanations of first, third and fifth grade children for certain phenomena, such as the disappearance of the sun during the night, the disappearance of stars during the day, the apparent movement of the moon, and the alteration of day and night. How do the perceptions change?
Methods: Dividing the participants into three groups according to the grades and collect data respectively
- 3 **Objective:** To present the misconceptions of school-age children about issues related to global warming / global climatic change
Research questions: What are student conceptions of the causes of global warming? How to improve their understanding of the issues?
Methods: Questionnaire and interviews

Reading an abstract

TASK

Abstract 1

- 1 This study is an attempt to contribute to the growing body of knowledge about students' conceptions and views concerning environmental and natural resource issues.
- 2 The method was to give students written tasks of the open-ended type.
- 3 The students' responses also indicate that they do not fully understand what fundamental societal changes would occur as a result of a drastic reduction in CO₂ emission. On the other hand, they are rather well

2

informed about how injurious depletion of the ozone layer is to humans.

Abstract 2 This paper reports on research concerning student teachers' perceptions about aspects of climate change as well as about greenhouse effect and ozone layer depletion, in order to take findings into account for teaching student teachers about these issues.

- 2 An open-ended questionnaire was used in order to gain a more comprehensive understanding of their thought.
- 3 These students believe that climate change is under way and base their beliefs on their own experience. They are unaware of the proper actions to be taken for slowing down the climate change, they also hold the misconception that ozone depletion, acid rain, and pollution in general are conducive to climate change. They confuse greenhouse effect with ozone depletion as far as the mechanisms through which they occur are concerned.

Identifying topic sentences for a general idea

TASK 1

Para. 21 Overall, students show increased understanding of the target phenomena, but there is room for growth

Para. 22 While there is certainly room for growth, students do show that they learned more about the components involved in the greenhouse effect.

Para. 23 Once they acquire new information they modify their ideas to be more scientifically valid.

Para. 24 As they reconcile new ideas with their pre-existing knowledge, students have generated an intermediate representation of the greenhouse effect that includes correct and incorrect ideas.

Para. 25 We will design more reliable methods to measure their understanding of the energy transformation that is a key to a deep understanding of this complex phenomenon.

TASK 2

Para. 1 Minimal efforts made to reduce the release of greenhouse gases, specifically carbon dioxide from the burning of fossil fuels, increase the rate and intensity of global warming.

Para. 2 The most prominent effect of global warming exists in the climate change over the past years.

Para. 3 Another area affected by global warming is the animal kingdom and nature.

Para. 4 A third type of evidence of global warming can be found in plants.

Para. 5 Global warming affects not only nature, but people and the economy.

Para. 6 Scientists directly link disease to global warming.

Para. 7 When people are trying to get others involved in the cause, a common problem that arises is the lack of motivation.

Para. 8 In conclusion, the world needs to put forth a stronger effort to prevent global warming.

Looking for visual and aural sources

Paying attention to the introduction

TASK 1

Topic of the lecture: ethical behavior on the Net

The structure of the lecture: 1) general ethical behavior; 2) ethical behavior in the use of computers

Topic of the lecture: Learning difficulty

The structure of the lecture: 1) the causes of learning problems; 2) a doctor's explanation

TASK 2

1 A 2 C 3 B 4 D 5 C

Understanding the ideas through examples

TASK

- 1 To introduce his topic and raise the awareness about the seriousness of global warming.
- 2 One is to illustrate the power of companies in addressing global warming and the other is to illustrate the role of individuals in this matter.
- 3 To illustrate the importance of policy in reducing CO₂.

- 4 The potential of radical innovation.
- 5 Reduce emissions in your home; be a green consumer; consider making your business carbon neutral; and become a catalyst of change.

Following the lecture through signal language

TASK

So let's look at this fourth factor—this is going to be a key one—and this is the amount of CO₂ put out per each unit of energy. And so the question is, can you actually get that to zero? If you burn coal, no. If you burn natural gas, no.... Now, I thought, how could I really capture this? Is there some kind of natural illustration, some demonstration that would grab people's imagination here? I thought back to a year ago when I brought mosquitos, and somehow people enjoyed that. *(Laughter)* It really got them involved in the idea of, you know, are there people who live with mosquitos? So, with energy, all I could come up with is this?... Let's look first at the burning fossil fuels, either burning coal or burning natural gas. What you need to do there, seems like it might be simple, but it's not, and that's to take all the CO₂, after you've burned it, going out the flue, pressurize it, create a liquid, put it somewhere, and hope it stays there.... Next, would be nuclear. It also has three big problems. Cost, particularly in highly regulated countries, is high. The issue of the safety, really feeling good about nothing could go wrong, that, even though you have these human operators, that the fuel doesn't get used for weapons. And then what do you do with the waste? And, although it's not very large, and there're a lot of concerns about that people need to feel good about it. So three very tough problems that might be solvable, and so, should be worked on.... And, finally, this storage problem. And, to dimensionalize this, I went through and looked at all the types of batteries that get made, for cars, for computers, for phones, for flashlights, for everything, and compared that to the amount of electrical energy the world uses, and what I found is that all the batteries we make now could store less than 10 minutes of all the energy.... Now, how we're gonna go forward on this: What's the right approach? Is it a Manhattan project? What's the thing that can get us there? Well, we need lots of companies working on this, hundreds. In each of these five paths, we need at least a hundred people. So, what do we have to do? What am I appealing to you to step forward and drive? We need to go for more research funding. You know, when countries get together in places like Copenhagen, they shouldn't just discuss the CO₂.

3 Literacy Skills

Paraphrasing

Dealing with the shared language

TASK 1

- 1 b, a, c
- 2 a, b, c
- 3 c, a, b
- 4 b, a, c
- 5 a, b, c

TASK 2

- ₅ By using synonyms
- ₄ By using the passive voice or active voice
- ₁ By using different order
- ₃ By using the negative or positive structure
- ₆ By using impersonal structures
- ₂ By using different parts of speech

TASK 3

- 1 The tourist industry in places like Hawii, which is fairly remote from other countries, has been adversely affected by rises in airfare.
- 2 Research has shown that having certain blood types increases susceptibility to certain diseases, but no such proof exists showing a relationship between blood types and people's personalities.
- 3 To detect and ultimately abort female fetuses, certain societies use ultrasound devices because of their low cost and portability.
- 4 Africa, which in 2011 became second only to Asia in number of cell phone users, is expanding its cellular capabilities to adequately connect the 600 million who have cell phones.
- 5 A shortage of fresh water to supply the billions in the world is apparently serious enough that it could eventually lead to wars. Among the factors contributing to the drop in fresh water supplies are issues related to the increasing population, sanitation, and environment.

Author-focused version and information-focused version

TASK

- 1 According to Jacqueline K. et al. (2008), Faculty and administrators need to understand three issues: (1) there are many factors for academic dishonesty, (2) there is no basic difference between plagiarism and cheating, and (3) compared with others, students do not regard plagiarism and cheating in academic studies as a moral problem.
- 2 Instructivists hold that the "real world," external to individuals, can be represented as knowledge and determines what will be understood by individuals. This view has been shifting to a constructivist view over the past decade (Merriënboer, 1997).
- 3 Two components must be present in an instructional design theory. The first component (methods) describes how human learning will be supported, and the second component (situation) describes when certain methods ought to be used (Reigeluth, 1999).
- 4 Man has always had an interest in the environment both as a source of raw materials and as a refuge for the human spirit". Nowadays, the two main environmental interests are based on the concept of a "better quality of life", as well as the need to replenish the sources of raw materials. In comparison with the pre-1960s, much greater interest in the environment is currently being expressed.
- 5 According to Gredler (2001), the same factors apply to developing complex skills in a classroom setting as to developing complex skills in any setting. A response must be induced, then reinforced as it gets closer to the desired behavior. Reinforcers have to be scheduled carefully, and cues have to be withdrawn gradually so that the new behaviors can be transferred and maintained.

Unit 3 Writing a Literature Review and an Introduction

1 Critical Reading

Understanding the text

TASK 1

Key words	GM soya, rat pups, influence
Objective	to see the effect of GM crop on the birth rate and mortality of rat pups
Research questions	How does GM soya affect the birth weight and survival of rat pups?
Methods used	experiment, with the experimental group and the control group

TASK 2

- 1 There is a lack of investigations on the influence of GM crops on mammals, especially on their reproductive function.
- 2 In order to see the effect of different diets on pregnancy, lactation and the growth of the rat pups, the group of GM soya and the group of traditional soya should be compared with the positive group whose rats received no soya at all.
- 3 Compared with other groups, the GM soya group had a very high rate of pup mortality and its pups continued to die over the period of lactation. The weight of their bodies and internal organs was lower.
- 4 The insertion of the foreign genes into soya and the accumulation of Roundup residues in GM soya might explain the negative effect.

TASK 3

Paras.1-2 1.e 2.f 3.c 4.g 5.a 6.d 7.h 8.b 9.i 10.j

Paras.3-5 1.g 2.f 3.d 4.h 5.k 6.i 7.a 8.e 9.j 10.c 11.b

Paras.6-11 1.c 2.a 3.e 4.h 5.i 6.b 7.g 8.d 9. f

Paras.12-13 1.d 2.c 3.e 4.a 5.b

TASK 5

- 1 Therefore, in order to have a better understanding of the effect of GM crop on the birth rate, mortality and weight gain of rat pups, we decided to conduct an experiment by feeding their mothers a kind of GM foods before their birth.
- 2 The pups from the GM group showed some changes in their internal organs with their growth. The traditional soya, however, did not show their negative effects.
- 3 The individual pups should be fed more milk since they were smaller. The consumption of the milk which contains GM soya flour might deprive them of a better chance to grow healthily.

TASK 7

Investigation of the influence of GM soya on the birthrate and survival of the offspring of Wistar rats was performed. A group of female rats were fed with GM soya flour before mating and pregnancy. The control group of females were fed with traditional soya and the third group of females, the positive control group, received feed without any soya. The weight and the mortality rate of the newborn pups were analyzed. The study showed that there was a very high rate of pup mortality (55.6%) in the GM soya group in comparison with the traditional soya group and the positive control group (9% and 6.8% respectively). Moreover, death in the first group continued during lactation, and the weights of the survivors were lower than those from the other two groups. It was revealed in these experiments that GM soya could have a negative influence on the offspring of Wistar rats.

Enhancing language ability

TASK 2

- 1 novel, adverse
- 2 established, stunt
- 3 hazard, detrimental
- 4 allocated, are / were immune to
- 5 supplement
- 6 vitally, persist
- 7 undertaken, unintended
- 8 presumed, a wide/widespread/broad distribution
- 9 mortality
- 10 speculate, transferred

TASK 3

- 1 **stunt:** prevent, restrict, hold up, hinder, hamper, impede, curb
- 2 **scheme:** plan, strategy, program, design, proposal
- 3 **novel:** different, new, unusual, altered, fresh, uncommon
- 4 **objective:** target, aim, purpose, goal, mission
- 5 **allocate:** give, assign, dispense, distribute
- 6 **show:** betray, suggest, demonstrate, indicate
- 7 **accumulate:** increase, gather, build up, grow, pile up, amass
- 8 **presume:** assume, postulate, believe, suppose, surmise
- 9 **detrimental:** adverse, destructive, pernicious, harmful, damaging, unfavourable
- 10 **perform:** conduct, implement, launch, institute, undertake, execute

Doing research projects

TASK 1

- 1 How many people try to avoid eating GM foods?
Because he wanted to show that people have a poor knowledge of what GM foods are.
- 2 The contamination caused the sickness because only the people consuming the supplement produced from Showa Denko KK company got the sickness.
- 3 They genetically engineered the bacteria during the production of L-tryptophan.

- 4 The company changed the filtration process during the manufacture which may bring more contamination to get through the products.
- 5 Because people got sick before filter was changed. Actually, the epidemiology has been prevalent since 1974 when the genetic engineered bacteria were introduced.

2 Academic Writing

Understanding the literature review

TASK 1

- 1 Paras. 2 and 3.
- 2 About 10 studies.
- 3 The hazard of genetically modified organisms (GMO) has been shown on animals.
- 4 There is a lack of investigations on the influence of GM crops on mammals, especially on their reproductive function.
- 5 Experimental researches in mice showed that ingested foreign DNA can persist in fragmented form in the gastrointestinal tract, penetrate the intestinal wall, and reach the nuclei of leukocytes, spleen and liver cells. Foreign DNA, orally ingested by pregnant mice, was discovered in blood (leukocytes), spleen, liver, heart, brain, testes and other organs of fetuses and newborn animals.

TASK 2

- 1 Because the author wanted to report previous studies in detail.
- 2 A thematic literature review.
- 3 Four.
- 4 Subsection 1 reports the general consumer attitudes towards GM Foods. Subsection 2 is about European consumer attitudes. Subsection 3 tells Chinese consumer attitudes. Last subsection summarizes gatekeepers' attitudes.
- 5 Subsection 1 has about 12 sources. Subsection 2 has about five. Subsection 3 has about three. Subsection 4 has about five.
- 6 It contains more details than the literature review of the Critical Reading text.

Writing a good literature review

TASK 1

- 1 No. It gives definition of GMOs.
- 2 Yes. Para. 2 tells the previous studies about general harmful effects of GM Foods on animals. Para. 3 demonstrates the effects on mice.
- 3 The introduction section of "The Effects of Food Deprivation on Concentration and Perseverance" is similar to that of the Critical Reading text in structure.
Para.1 introduces the topic and general information. Para. 2 reports the early studies. Para. 3 analyzes the limitations of the previous studies. The last paragraph points out the significance of the current study.

TASK 2

Attitudes towards GM foods vary from country to country. In Sweden, for example, people generally hold a negative view from the moral and ethical perspective. Some of them would buy

GM foods simply because of their environmental effects and nutritious aspects rather than low prices and good taste (Magnusson, et al., 2002). In Norway, however, people would purchase GM foods if they have a greater discount though they generally prefer to eat traditional foods which are safer. It seems that formal education about biotechnology plays an important role in their unwillingness to accept GM foods (Kristine, et al., 2002). Similarly, in China, people's attitude is not clear-cut according to Knight and Gao (2009). Most Chinese consumers, though concerned about the safety of GM foods, would accept them out of the price benefits.

Using appropriate tense in literature review

TASK

- 1 indicated / have indicated; is; identified; tend; will be
- 2 is; are / were; was; were; indicated

Writing an introduction section

TASK 1

- 1 Both introduce the background information and the objective studies.
- 2 The introduction of the Critical Reading text incorporates its literature review into the introduction section, while this introduction here is without literature review.
- 3 The background information can be found in the first 3-4 sentences of the introduction sections of both articles.
- 4 The last sentence of the Critical Reading text indicates its objective, while the objective of the text here can be found in the last two sentences.
- 5 The Critical Reading text incorporates its literature review into the introduction section, while the text here has a separate literature review section.

TASK 2

Academic Dishonesty Among College Students

Introduction

Academic dishonesty is a problem that has been plaguing colleges and universities for generations. An investigation of any institution today will certainly reveal some form of academic dishonesty. [an introduction of the topic and its background] Researchers of academic dishonesty vary in their reports of how many students cheat in college. For example, a survey conducted by Whitley (1998) revealed that an average of 70.4% of students have admitted to cheating in some form during their college careers. Numerous studies have pointed out that men are more likely to cheat than women (Dawkins, 2004; Mustaine & Tewksbury, 2005). Younger students have been found to be more likely to cheat than older students (Jensen et al., 2002), but findings concerning class standing do not always follow a linear pattern (Whitley, 1998). The activities in which a student is involved play a major role in determining the level of academic dishonesty. Cheating has been found to be more common among students who are enrolled in more credit hours, student athletes, members of social fraternities (Mustaine & Tewksbury, 2005), and students who work fewer hours per week (Whitley, 1998).

With so many students participating in academic dishonesty, one might expect other students to know about it as well. Not surprisingly, 71% of students report having witnessed others cheating

at least once (Dawkins, 2004). However, out of this group only 1% are willing to tell their professors when they see others cheating (Jensen et al., 2002). When looking at the differences between motivations for cheaters and non-cheaters, Jordan (2001) found that cheaters were extrinsically motivated in the classes in which they cheated while non-cheaters were motivated by the mastery of the course. [a review about the previous studies]

However, most research on academic dishonesty has relied primarily on self-reports of cheating behaviors. [the limitation of the previous studies] The present study includes an empirical portion in which participants are put in a situation in which cheating may be to their advantage (see Methods section for a full explanation). [the methodology] I hypothesized that participants would be most likely to cheat when they are offered a monetary reward for success. [the hypothesis of the author] The purpose of the study is to investigate participants' attitudes toward cheating and the effects of academic motivation, self-efficacy (自我效能), and academic integrity on cheating behaviors. [the objective the paper]

The Effects of Food Deprivation on Concentration and Perseverance

Introduction

Many things interrupt people's ability to focus on a task: distractions, headaches, noisy environments, and even psychological disorders. To some extent, people can control the environmental factors that make it difficult to focus. However, what about internal factors, such as an empty stomach? Can people increase their ability to focus simply by eating regularly? [an introduction of the topic and its background]

The earliest investigation of the general effects of food deprivation found that long-term food deprivation (36 hours and longer) was associated with sluggishness, depression, irritability, reduced heart rate, and inability to concentrate (Keys, Brozek, Henschel, Mickelsen & Taylor, 1950). Another study found that fasting for several days produced muscular weakness, irritability, and apathy or depression (Kollar, Slater, Palmer, Docter & Mandell, 1964). Since that time, research has focused mainly on how nutrition affects cognition. However, as Green, Elliman, and Rogers (1995) point out, the effects of food deprivation on cognition have received comparatively less attention in recent years.

The relatively sparse research on food deprivation has left room for further research. First, much of the research has focused either on chronic starvation at one end of the continuum or on missing a single meal at the other end (Green et al., 1995). Second, some of the findings have been contradictory. One study found that skipping breakfast impairs certain aspects of cognition, such as problem-solving abilities (Pollitt, Lewis, Garza & Shulman, 1983). However, other research by M. W. Green, N. A. Elliman, and P. J. Rogers (1995, 1997) has found that food deprivation ranging from missing a single meal to 24 hours without eating does not significantly impair cognition. Third, not all groups of people have been sufficiently studied. Studies have been done on 9-11-year-olds (Pollitt et al., 1983), obese subjects (Crumpton, Wine & Drenick, 1966), college-age men and women (Green et al., 1995, 1996, 1997), and middle-aged males (Kollar et al., 1964). Fourth, not all cognitive aspects have been studied. In 1995 Green, Elliman, and Rogers studied sustained attention, simple reaction time, and immediate memory; in 1996 they studied

attentional bias; and in 1997 they studied simple reaction time, two-finger tapping, recognition memory, and free recall. In 1983, another study focused on reaction time and accuracy, intelligence quotient, and problem-solving (Pollitt et al., 1983). [a review about the previous studies]

According to some researchers, most of the results so far indicate that cognitive function is not affected significantly by short-term fasting (Green et al., 1995). However, this conclusion seems premature due to the relative lack of research on cognitive functions such as concentration and perseverance. To date, no study has tested perseverance, despite its importance in cognitive functioning. In fact, perseverance may be a better indicator than achievement tests in assessing growth in learning and thinking abilities, as perseverance helps in solving complex problems (Costa, 1984). Another study also recognized that perseverance, better learning techniques, and effort are cognitions worth studying (D'Agostino, 1996). Testing as many aspects of cognition as possible is a key because the nature of the task is important when interpreting the link between food deprivation and cognitive performance (Smith & Kendrick, 1992). [a statement of the previous studies' limitation]

Therefore, the current study helps us understand how short-term food deprivation affects concentration on and perseverance with a difficult task. Specifically, participants deprived of food for 24 hours were expected to perform worse on a concentration test and a perseverance task than those deprived for 12 hours, who in turn were predicted to perform worse than those who were not deprived of food.

TASK 3

1 A 2 B 3 E 4 C 5 D 6 F

3 Literacy Skills

Writing a good synthesis

TASK 1

- 1 A number of conclusions based on research into the effects of violence in computer games seem to be flawed. Walker's findings (2001, p.116) show that "young people who regularly play violent computer games have a tendency to be more aggressive than those who do not". If this research compares children who have been equally exposed to violence in other media, Smith's (2003) arguments about violence in the media are not relevant. In addition, Walker only examined violent computer games, so Smith's argument about non-violent computer games is irrelevant.

- 3 Students need practical strategies to check the reliability of information on websites. Chan (2001) suggests three criteria for judging whether the information on a website is reliable. These are

authorship, status of the website, and credibility of the document itself. Smith (2003) believes that it is too time-consuming to examine a website for these criteria, and that the criteria are therefore impractical. As Smith does not suggest an alternative, I believe that Chan's criteria should be used, but the extent of checking the criteria should be balanced with the time available.

TASK 2

Environmental policies play an important role in technological innovation in renewable energy (Johnstone et al., 2010). It is well known that traditional fossil fuels have suffered great disadvantages, but developing renewable energy technologies is costly and needs the government's financial support (Rana, 2011). Therefore, the government's policies are important to encourage innovation on sustainable technologies of renewable energy which can compete with traditional fossil fuels (Johnstone et al., 2010). This requires the better understanding of policy makers who can change their ideas and support the innovation on renewable technologies if they know how innovation occurs (Foxona et al., 2005)

References

- Foxona, T.J., R. Gross, A. Chase, J. Howes, A. Arnall, D. Anderson. 2005. "UK innovation systems for new and renewable energy technologies: drivers, barriers and systems failures" *Energy Policy*. 33(16): 2123–2137.
- Nick Johnstone, Ivan Haščič, David Popp. 2008. "Renewable Energy Policies and Technological Innovation: Evidence Based on Patent Counts" *Environmental and Resource Economics*. 45(1): 133-155.
- Vishwajeet Rana. 2011. The Energy Economics of Financial Structuring for Renewable Energy Projects. A dissertation submitted to the faculty of the Stevens Institute of Technology, UMI Number: 3467236.

Using multiple sources

Reporting Verbs

TASK 3

- 1 insist(ed) / argue(d)
- 2 described / regarded
- 3 suggests (suggested) / reveals (revealed)
- 4 note(d) / believe(d)
- 5 interprets (interpreted)
- 6 assumes (assumed)
- 7 suggested



学术英语理工（第二版）

Unit 4 Writing Methods and Results Sections

1

Critical Reading

Understanding the text

TASK 1

Key words	Infant-feeding practices, impacts, urbanization and socioeconomic status
Objective	to study the impact of urbanization and women's socioeconomic status on the change in infant-feeding practices
Research questions	What impacts do urbanization and women's socioeconomic status have on the change in infant-feeding practices and why there is a drastic change in traditional breast-feeding practices?
Methods used	questionnaires and interviews

TASK 2

- 1 This area is heterogeneous in its ethnic and socioeconomic composition and it is widely assumed that women there suffer negative effects of urbanization and modernization on infant-feeding practices and the traditional African image of mothering.
- 2 Most of these earlier reports attributed the steady decline to attitudes and influences imported from the developed world and the preference for bottle-feeding by the urbanized African.
- 3 To them urban life means extreme individual and community poverty. It exerts a great pressure on mothers to spend more time trying to augment the family income at the expense of caring adequately for their children.
- 4 The main reason is the poverty of urban people rather than the ignorance on the mother's part.

TASK 3

Para. 1

- 1 ☐ 2 vegetative 3 ☐ 4 urbanize 5 importance
6 ☐ 7 indefinitely 8 currency, occurrence 9 explain

Para. 2

- 1 protective, paternal 2 ☐ 3 ☐ 4 correctable
5 different, dissimilar 6 ☐ 7 caring for

Paras. 3-9

- 1 augment 2 secondary 3 clinical practice 4 describe 5 portion
6 substance, allowance 7 ☐ 8 delayed 9 domestic

Paras. 10-11

- 1 ☐ 2 take notice 3 interlink 4 ☐
5 contain 6 ☐ 7 innumerate 8 ☐

Paras. 12-13

- 1 retrain 2 substitute 3 travel 4 ☐ 5 expressed 6 ☐
7 ☐ 8 ☐ 9 present 10 originality 11 ☐

Paras. 14-15

- 1 unbalanced diet 2 ☐ 3 ☐ 4 ☐ 5 ☐
6 change 7 arrangement 8 ☐ 9 divergence 10 citizen

TASK 4

Paras. 1-5	3	5	2	1	4
Paras. 6-10	8	7	10	6	9
Paras. 11-15	14	12	11	15	13

TASK 5

- 1 Monthly incomes of almost 29% of the families are below N 200, which could barely support the basic necessities of urban life in a city where housing price and the cost of life is one of the highest in the world.
- 2 The fact that the actual practice did not match the mother's nutritional awareness failed to support the frequently cited argument that the ill-educated urban women did not receive proper advice.
- 3 Obviously such explanations can't be applied to many situations where it was the poverty of the urban mothers that made it difficult for them to breastfeed their children as long as they expected.

TASK 6

Breast-feeding could effectively protect women from breast cancer.

TASK 7

Breast-feeding, a traditional infant-feeding practice in Nigeria, has undergone a serious decline recently. In order to explore the factors responsible for this negative change and offer some suggestions based on the result, a survey was conducted among Nigerian women resident in Surulere. The infant-feeding practice and attitudes of 558 mothers were examined via questionnaires and personal interviews. The result showed that despite the women's awareness of the importance of breast-feeding practice, which many of them believed should continue for up to one year, only about 24.6% of the mothers studied practiced breast-feeding for that long and 81% used supplementary foods for infants from birth to three months. It is suggested that economic and social pressures were responsible for the change in traditional infant-feeding practices. Hence unless the poverty of urban women is solved, any nutrition education program will be meaningless.

Enhancing language ability

TASK 2

- 1 erroneously, identified
- 2 critical, prestige
- 3 attributed, discrepancies
- 4 prevalence, reversed
- 5 address, assume
- 6 underscored, augment
- 7 impoverished, instituted
- 8 unanimity, extensive
- 9 retain, synonymous
- 10 eliminated, heterogeneous

TASK 3

- 1 ☐ 2 comment 3 student 4 ☐ 5 ☐
6 ☐ 7 ☐ 8 ☐ 9 ☐

Doing research projects

TASK 1

- 1 Is there a casual relationship between breast feeding and social mobility?
- 2 They obtained the data from the British Birth Cohort studies from 1946 to 2001.
- 3 They used two different ways: to replicate the results from two different studies and to use the statistical methods mimic the randomized.
- 4 First, the breast feeding rate in 1958 was much higher than 1970, Second, the difference between the social grade was not so much great in 1958 than 1978 cohort.
- 5 Although the patterns were different in two cohorts, the interpretation of the casual between breastfeeding and social mobility was consistent.

2 Academic Writing

Writing a methods section

TASK 1

- 1 An interview.
- 2
 - ethnic origin of the mothers
 - level of education
 - occupation and age of the mothers
 - total income of the family
 - number of children born alive to the mothers
 - type of infant feeding preferred by the mothers and factors influencing the decision
 - age at which breast-feeding was supplemented with other foods
 - specific types of supplementary foods preferred by the mothers
 - duration of breast-feeding
 - influence of hospitals / clinics and advertisers of baby foods on choice of infant-feeding practice.
- 3 The research questions are: 1) What are current maternal perceptions of infant-feeding among different ethnic, religious, cultural, and socioeconomic groups in Nigeria ? 2) Is there a decline in breast-feeding ? 3) What are major factors responsible for the negative change ? 4) How can we institute corrective educational measures ?
- 4 Yes. But a questionnaire can also be used. The participants can include doctors and nurses in hospitals.

TASK 2

1

Text	Questionnaire	Interview	experiments
in Unit 1	✓		
in Unit 2			✓
in Unit 3			✓
in Unit 4		✓	
below			✓

2 Three. They are questionnaire, interview and experiments.

3 To find the effect of GM foods on animals, the method of experiments is most appropriate. The text in Unit 2 uses the same method to evaluate the change in the students' perceptions and knowledge about greenhouse effects. Questionnaire is an appropriate method according to the research questions of the Critical Reading text in Unit 1.

4 The detailed procedures allow the reader to duplicate the experiment if such were desired to confirm, or refute, the author's findings.

TASK 3

- 1 Y Y N
- 2 Y Y Y
- 3 Y Y N
- 4 Y Y Y

TASK 4

- 1 Y Y N
- 2 N N Y
- 3 Y Y N

TASK 5

- 1 Y Y N Y
- 2 Y Y Y Y

Controlling variables

TASK 1

4 the age and physical condition of the subjects in each group
5 the size of the cages

4 exposure to mass media
5 the number of children
6 income
7 religions
8 the access to free samples

TASK 2

Content	Methods 1	Methods 2	Methods 3
Research questions	1) What are current maternal perceptions of infant-feeding among different ethnic, religious, cultural, and socio-economic groups in Nigeria? 2) What are major factors responsible for the negative change? 3) How can we institute corrective educational measures?	What are their perceptions of GM foods (such as benefits, risk, trust, acceptability)?	What are the gatekeepers' perceptions of GM technology?
Methods	Interview & Questionnaire	Questionnaire	Interview
Participants / subjects	Nigerian mothers	The households in Norwich	Gatekeepers such as company leaders in the commercial centers in China
Procedure	Individual interview	Distributed to the households	Personal interview for 1 hr. to 2hrs. tap recording
Variables	Education, occupation, age, income, religions	Age, education, exposure to mass media, price	Their interest, government policy, the customers' acceptability

Writing a results section

TASK 1

- 1 Both use tables and figure to report the results.
- 2 Both are clear. But Unit 4 uses a topic sentence to summarize each result in each paragraph. Hence it is more readable.
- 3 Para. 9.
- 4 Paras. 7-10.
- 5 Both use phrases.

Using tables and figures effectively

TASK 1

- 6 Unit 3 uses 5 tables, and Unit 4 uses 10.
- 7 No, the authors select some most related data to report.
- 8 Because not all the data are closely related to the research questions.
- 9 It depends.

- 10 Phrases.
- 11 Tables take less space than charts and more accurate.

Using correct tenses in reporting results

TASK 2

shows, were, was, stopped, was, was, suggests, is

3 Literacy Skills

Writing in a formal style

TASK 1

Version A: How to make people work harder [Imprecise vocabulary – use “motivation?”] is a topic that lots of people [Vague – give names] have written about in the last few years [Vague – give dates]. There are lots of [Avoid “lots of?”] different theories and I think [Too personal] some of them are OK [Too informal]. When we think about this [Too personal] we should remember the old Chinese proverb that you can lead a horse to water [Do not quote proverbs] but you can’t make it drink. So how do we increase production? [Avoid rhetorical questions] It’s [Avoid contractions] quite a complex subject but I’ll just talk about a couple [Too personal and informal] of ideas.

TASK 3

- 1 If the environment is not protected, children will suffer the most.
- 2 These problems need to be address. / It is necessary to address these problems.
- 3 However, there are many restaurants serving endangered animals.
- 4 Participating in social activities as volunteers can be a beneficial / positive experience for children.
- 5 Economic situation will not improve until efforts are made by the government.
- 6 The investigation yielded no new results.
- 7 The government allocated little funding for the program.
- 8 This problem has few viable solutions.

The use of nominalization

TASK 2

- 1 The model provides an explanation for the fluctuations in the figure in this period.
- 2 The space skill has found application for the exploration of the universe.
- 3 The accidental release of radiation causes widespread long-term damage.
- 4 To ensure the widest acceptability and greatest profitability, the novel follows the most commonly accepted notions of morality and justice.
- 5 The richness of the soil in this area favored the growth of crops.
- 6 The explosion of information and the increasing obsolescence of knowledge is forcing upon higher education a new set of demands.
- 7 The signs of the times point to the necessity of the modification of the legal system.
- 8 Careful supervision is necessary to ensure the survival of the crops until harvest time.

- 9 Yet all of history has taught us that the denial of these ultimates and the placement of man at the core of the universe will result in mass selfishness.
- 10 A knowledge of its sources and symptoms helps explain many things otherwise impossible for rational analysis.

The use of passive voice

TASK 2

- 1 Full advantage must be taken of the opportunities offered by urbanization.
- 2 However, little control is exercised over online articles and speeches.
- 3 The main effects of poverty can be illustrated by looking at those on the lowest incomes in rich cities.
- 4 It's so obvious that jobs are given according to whether they were male or female.
- 5 Despite potential benefits of such techniques, serious concerns have been raised about the potential environmental and medical consequences of GMOs.
- 6 A fairly consistent picture of its short-term effects on users is presented in many publications.
- 7 Much remains to be learned about the acute physiological effects of Cannabis.
- 8 The participants were randomly assigned to three groups.
- 9 The distinction is made clear between government policy and economic reality.
- 10 Social conditions as well as the economic situation should be taken into account.

The use of single, formal academic verbs

TASK 2

- 1 assist
- 2 established; develop
- 3 increased
- 4 discovered
- 5 eliminate
- 6 investigating / examining
- 7 raised
- 8 observes; contributed to / caused
- 9 constitute
- 10 conduct
- 11 accelerated
- 12 caused
- 13 developed
- 14 encountered

TASK 3

- 1 solid
- 2 fatigued; lengthy
- 3 attempts; fundamental
- 4 confirms
- 5 potential

- 6 generates / yields
- 7 primary; distinguish
- 8 underlines
- 9 discrepancy
- 10 principal
- 11 considerable

Unit 5 Writing a Discussion Section and an Abstract

1

Critical Reading

Understanding the text

TASK 1

Key words	fraud in medical research, survey of biostatisticians
Objective	to investigate fraud in medical research
Research questions	What are the situation and characteristics of fraud in medical research? Why is it prevalent?
Methods used	questionnaires and interviews among biostatisticians

TASK 2

- 1 Because they routinely work closely with physicians and scientists in many branches of medical research and have unique insight into data. In addition, they have the methodological competence to detect fraud and could be expected to have a special professional interest in the validity of results. Biostatisticians therefore could provide unique and reliable information on the characteristics of fraud in medical research.
- 2 To ensure the validity and reliability of the data.
- 3 Career and power instead of financial reward.
- 4 Most responders have expertise and opportunity to contact data. They may judge critically scientific collaborators' optimistic interpretation of data.

TASK 3

Paras. 1-3

1-5 e c d i a

6-10 b f h j g

Paras. 4-8

1-7 f h a c m d b

8-14 k e g l i j n

Paras. 9-15

1-6 d g b i m k

7-13 h c l j e f a

TASK 4

1-5 3 2 1 4 5

6-10 9 10 8 7 6

11-15 13 15 14 12 11

TASK 5

- 1 It is difficult to investigate scientific fraud which is characterized by its secrecy and lack of reliable evidence. Therefore not many people are familiar with its characteristics, frequency and impact on medical research.
- 2 Biostatisticians cooperate frequently with physicians and scientists in many branches of medical research and have a unique opportunity to understand data. Furthermore, they have expertise to

discover fraud and interest in checking the validity of results.

- 3 As biostatisticians can access inside information and confidential data, they have a better understanding of the implications of data. Hence they know the detailed process before the result is published.

TASK 7

The paper is to conduct a survey of biostatisticians, who were members of the International Society for Clinical Biostatistics. The objective is to assess the characteristics of fraud in medical research. **[the objective]** The survey was performed between April and July 1998. The participation rate was only 37%. We report the results because a majority (51%) of the participants knew about fraudulent projects, and many did not know whether the organization they work for has a formal system for handling suspected fraud or not. Different forms of fraud (e.g., fabrication and falsification of data, deceptive reporting of results, suppression of data, and deceptive design or analysis) had been observed in fairly similar numbers. **[the methods used]** We conclude that fraud is not a negligible phenomenon in medical research, and that increased awareness of the forms in which it is expressed seems appropriate. **[the major findings]**

Enhancing language ability

TASK 2

- 1 implication, undermine
- 2 deceptive, episodes
- 3 confidential, disclose
- 4 in a position
- 5 Respondents, mislead
- 6 reiterated, intervene
- 7 endemic, fabricate
- 8 optimistic, are amenable to
- 9 motive, suppress
- 10 occurrences

TASK 3

- 1 **endemic:** widespread, rife, pervasive, common, sweeping, extensive, prevalent
- 2 **episode:** event, experience, happening, incident, occurrence
- 3 **detect:** ascertain, *uncover*, discover, track down
- 4 **motive:** purpose, intention, motivation, grounds, *reason*, cause
- 5 **borderline:** ambiguous, indefinite, doubtful, inexact, indeterminate, marginal, problematic
- 6 **disclose:** publish, reveal, unveil, expose, unmask, exhibit, uncover, make public
- 7 **deceptive:** treacherous, fraudulent, deceiving, unreliable, false, misleading, fake
- 8 **methodology:** method, techniques, approach
- 9 **explanation:** analysis, exposition, reason, interpretation
- 10 **undermine:** worsen, damage, compromise, harm, jeopardize, endanger

Critical thinking (examining evidence)

TASK 1

- 1 An investigation of an international society for biostatisticians because they could provide unique and reliable information on the characteristics of fraud in medical research.
- 2 A survey of scientists themselves who know more of scientific misconduct in their disciplines both at home and abroad.

TASK 2

- 1 Case examples
- 2 Statistic evidence from a survey
- 3 Examples
- 4 Quotations from the experts

TASK 3

The evidence is not reliable as both Dr. Worries and the association of cosmetic surgeons are commercially involved in the matter. Their claims are not objective.

Doing research projects

- 1 Misconduct was more tight and rigid definition which did not need define or interpretation.
- 2 It involves three areas such as fabrication, falsification and plagiarism.
- 3 Citing previous works, accessing and sharing data, and collaborative and authorship.
- 4 There is growing attention paid to the ethics problems.

2 Academic Writing

Writing a discussion section

TASK 1

Contents	Text I	Text II
Major findings	Para. <u>9</u>	Para. <u>1</u>
Comparison with other studies	Para. <u>10,12,13</u>	Para. <u>2</u>
Explanations	Para. <u>11</u>	Para. <u>2</u>
Limitations or future direction	Para. <u>14</u>	Para. <u>3</u>
Implications or suggestions	Para. <u>15</u>	Para. <u>4</u>

TASK 2

Para. 2 Compare with the studies of Europe about fears

Para. 3 Suggestions

Para. 4 Major findings

Para. 5 Possible implications

Para. 6 Limitations and future directions

TASK 3

1-7 B C G D E F A

TASK 4

Academic Dishonesty Among College Students

Discussion (Introduction section P. 124)

- 1 I hypothesized that the participants would be most likely to cheat in the 20 dollar reward group. However, I did not find a significant difference between the different reward levels, which could either mean that the type of reward did not affect the likelihood of cheating or that the sample size was not large enough to yield significant results. [\[explain the unexpected result\]](#)
- 2 The pattern that emerged between participants' self-reports of cheating behaviors and their dishonesty ratings revealed that participants were willing to say that behaviors in which they had not engaged were "extremely wrong," but behaviors in which they had engaged were rated as "not wrong" or "somewhat wrong." It could be that the participants were trying to justify their behaviors by saying that they only engaged in the more acceptable types of cheating and never took part in anything that was really dishonest. [\[explain the result\]](#)
- 3 Participants who cheated on the puzzle task also had higher self-efficacy scores, which means that they feel like they have more control over their success. Perhaps participants who have higher levels of self-efficacy are more confident that they will succeed. The puzzle task in this experiment, however, was designed so that the participants could not successfully solve all of the puzzles without cheating. Refusing to fail such a simple task, they decided cheating was an acceptable tactic for success. [\[explain the result\]](#)
- 4 This study contained only 34 participants, hence it is important to note that all of these findings should be approached with caution. Further investigation is necessary in order to strengthen the significance of the findings. However, the findings of this study could be the first steps to uncovering the motivations behind academic dishonesty. A suggestion for future research is closer examination of the differences in academic motivations of people who cheat and those who do not. [\[limitations and future research\]](#)

Health-related Quality of Life and Health Risk Behaviors Among Smokers

Discussion

- 1 This is the first national population-based study of U.S. adults examining associations among smoking status, HRQOL (Health-related Quality of Life), and health risk behaviors. The relationship between smoking and impaired physical health is corroborated by the results of this study. Specifically, compared to persons who never smoked, current smokers are more likely to report fairly poor general health, frequent physical distress, frequent activity limitations, and frequent pain. [\[Major findings\]](#) While previous research suggests that people with poor mental health are more likely to smoke than those without, few previous studies have examined whether those who smoke are more likely than those who do not to report poor mental health. Additionally, previous research in this area has yielded inconsistent results. In particular, while some prior studies reported mental health impairments among smokers, others found associations in women but not men or found no association. These results suggest that there is a significant association between smoking and impaired mental health among both men and women residing in the

community. Specifically, those who currently smoked were significantly more likely than those who formerly smoked or never smoked to report frequent mental distress, frequent depressive symptoms, frequent anxiety symptoms, and infrequent vitality. [Comparison with previous studies]

- 2 Respondents who currently smoked were more likely than former smokers or never smokers to be heavy or binge drinkers, but were less likely to be obese. It is well known that smoking is frequently associated with alcohol consumption, and that smoking cessation is often associated with weight gain. Research suggests that fears about weight gain may motivate the initiation or continuation of smoking, especially among women. However, cognitive behavioral therapy techniques have been demonstrated to improve smoking cessation rates among smokers concerned with gaining weight.
- 3 Most notably, these results reveal that in addition to being associated with increased alcohol consumption and affective disturbance, smoking among community dwellers is also associated with physical inactivity, sleep impairment, and inadequate consumption of fruit and vegetables. Thus, smoking appears to be associated with a constellation of health behaviors, potentially further compounding its adverse effects. As physical inactivity, inadequate sleep, and low consumption of fruit and vegetables have each been associated with increased morbidity and mortality, the presence of smoking should alert physicians to the possible coexistence of other adverse health behaviors that may compound its deleterious consequences. In short, smoking may represent the “tip of the iceberg” of a number of modifiable risk factors meriting intervention. [Explanation of the result]
- 4 This study has several limitations. First, it is difficult to estimate the association between HRQOL and smoking without assessing physical and psychiatric comorbidities (并发症). Because the survey did not assess medical comorbidities, adjustments could not be made for these factors. Second, because the study was cross-sectional, causality cannot be inferred. Nonetheless, these findings demonstrate a strong association between smoking and HRQOL, suggesting that clinicians should be alert to the potential presence of psychiatric comorbidities among their patients who smoke as well as to the presence of smoking among patients with psychiatric disorders. Third, these analyses are based on self-reported data, and therefore these results could be influenced by reporting biases. However, the Behavioral Risk Factor Surveillance System (BRFSS) data on cigarette smoking measures have moderate to high validity and high reliability, and self-reports of smoking have been found to be valid in other population based surveys. Fourth, because the analysis was based on data from 23 states and the District of Columbia, the results may, conceivably, not be representative of the entire United States. Finally, while psychiatric symptoms were assessed, there was no assurance that these symptoms necessarily met diagnostic criteria for affective or anxiety disorders.
- 5 These results indicate that in addition to implementing smoking cessation efforts among patients who smoke, physicians should view smoking as a marker of a potential array of health risk behaviors. The results of this investigation also suggest that physical activity, sleep, and dietary characteristics may be important areas for physician assessment and, potentially, intervention among patients who smoke. [Implication]

TASK 5

1-5 D C B A E F

Writing an abstract

TASK 2

Academic Dishonesty Among College Students

Abstract

The present study investigated the effects of academic motivation, academic integrity, attitude toward cheating, and self-efficacy on cheating behaviors of college students. [the objective] Participants completed a puzzle-solving task in which some puzzles were not solvable unless a participant cheated. One group was offered a 20 dollar reward for the highest score, the second group's scores were made public, and the third group served as the control. [methods employed] There was not a significant difference in the cheating behaviors of the three groups. A significant negative correlation was found between participants' self-reports of cheating behaviors and how dishonest they rated those behaviors. Participants who cheated on the puzzle task had significantly higher self-efficacy scores than those who did not cheat. Implications for these findings are discussed. [major findings]

Laypeople's Understanding of Radioactivity and Radiation

Abstract

It may be argued that the population should have sufficient understanding of radiation phenomena to secure individual safety as well as democratic decisions. To attain this, it is necessary for the communicators of radiation information to be familiar with the lay person's perceptions of these phenomena. [the objective] A questionnaire survey was conducted to examine non-experts' conceptions of radiation phenomena. [the method] The survey revealed incomplete understanding of concepts such as radioactive decay, half-life and absorption of radiation and a lack of differentiation of between radiation and radioactive material. Many respondents did not distinguish between sources of ionizing radiation and other environmental hazards. Nuclear power plants and submarines were the most feared sources of radiation. There are indications that the lay understanding of radiation phenomena and risk is to a large extent formed by mass media and that "school knowledge" of these phenomena is not applied in situations belonging to the "real world". [major findings] To resolve this problem, new teaching and information procedures are needed. These should take into account the learners' perceptions and should integrate 'school knowledge' with considerations belonging to the "real world". [the suggestion]

Using Interactive Technology to Support Students' Understanding of the Greenhouse Effect and Global Warming

Abstract

In this work, we examine middle school students' understanding of the greenhouse effect and global warming. [the objective] We designed and refined a technology-enhanced curriculum module called *Global Warming: Virtual Earth*. In the module activities, students conduct virtual experiments with a visualization of the greenhouse effect. They analyze data and draw conclusions about how individual variables effect changes in the Earth's temperature. They also carry out inquiry activities to make connections between scientific processes, the socio-scientific issues, and ideas presented in the media.

[the method]Results show that participating in the unit increases students' understanding of the science. We discuss how students integrate their ideas about global climate change as a result of using virtual experiments that allow them to explore meaningful complexities of the climate system. [major findings]

Influence of Genetically Modified Soya on the Birth Weight and Survival of Rat Pups

Abstract

Investigation of the influence of GM soya on the birthrate and survival of the offspring of Wistar rats were performed in this paper. [the objective] A group of female rats were fed GM soya flour before mating and pregnancy. The control group of females were fed traditional soya and the third group of females, the positive control group, received feed without any soya. The weight and the mortality rate of the newborn pups were analyzed. [the methods employed]The study showed that there was a very high rate of pup mortality (55.6%) in the GM soya group in comparison with the control group and the positive control group (9% and 6.8% respectively). Moreover, death in the first group continued during lactation, and the weights of the survivors are lower those from the other two groups. [findings] It was revealed in these experiments, that GM soya could have a negative influence on the offsprings of Wistar rats. [the conclusion]

The Impact of Urbanization and Socioeconomic Status on Infant-feeding Practices in Lagos, Nigeria

Abstract

Breast-feeding, a traditional infant-feeding practice in Nigeria, has undergone a serious decline recently. [the background] In order to explore the factors responsible for the negative change and offer some suggestions based on the result, a survey was conducted among Nigerian women resident in Surulere [the objective]. The feeding practice and attitudes of 558 mothers were examined via questionnaires and personal interview. [the methods employed]The result showed that despite the women's awareness of the importance of breast-feeding practice which should continue for up to one year (38 %) according to their expressed opinion, only about 24.6 % of the mothers practiced breast-feeding for that long and 81 % used supplementary foods for infants from birth to three months. It is suggested that economic and social pressures were responsible for the change in traditional infant feeding practices. [findings]Hence unless the poverty of urban women was solved any nutrition education program was meaningless. [the conclusion]

Using the correct tense, voice and language in abstracts

TASK 1

The paper is to conduct a survey of biostatisticians, who were members of the International Society for Clinical Biostatistics. The objective is to assess the characteristics of fraud in medical research. [the objective] The survey was performed between April and July 1998. The participation rate was only 37%. We report the results because a majority (51%) of the participants knew about fraudulent projects, and many did not know whether the organization they work for has a formal system for handling suspected fraud or not. Different forms of fraud (e.g., fabrication and falsification of data, deceptive reporting of results, suppression of data, and deceptive design or analysis) had been observed in fairly similar numbers. [methods employed] We conclude that fraud is not a negligible phenomenon in medical research, and that increased awareness of the

forms in which it is expressed seems appropriate. **[the conclusion]** Further research, however, is needed to assess the prevalence of different types of fraud, as well as its impact on the validity of results published in the medical literature. **[future research]**

TASK 2

1-6 A D C F E B

3 Literacy Skills

Following the appropriate academic writing style

Using hedging language

TASK 1

Certainty Degree	Expressions
Strong modality	is, will, must always, never, definitely, certainly, clearly, obviously, unavoidably, it is certain/clear/obvious, to a great/large extent
Moderate modality	should, most would, can, tend to, usually, likely, probably, regularly, generally, normally, presumably, frequently, a significant proportion, numerous, it appears certain, it is usually the case that
Weak modality	May, might, could, possible, conceivable, occasionally, seldom, perhaps, maybe, uncertainly, somewhat, potentially, apparently, it is possible/unlikely that, it seems that, it may be the case that

TASK 4

- 1 In spite of its limitations, the study may have a number of important strengths.
- 2 It seems to prove that animals become old because, if they did not, there were no successive replacement of individuals and hence no evolution.
- 3 It suggests that the effect of these small particles is to cool the climate and to partially offset the warming of increasing concentrations of greenhouse gases.
- 4 Very high radiation doses might destroy body functions and lead to death within 60 days.
- 5 The effects of routine releases of radioactivity from nuclear plants tend to depend on how the spent fuel is handled.
- 6 Although they also began to eat the GM soya, the effect might be mediated by the two first factors.
- 7 It seems that all unethical behavior tends to be potentially criminal.
- 8 Mars rock might not be melted even at high temperatures.

Writing an end-of-text references list

TASK 1

One of the main problems tourists cause in Antarctica is the disruption of scientific research being conducted there. While scientific study in Antarctica was once the main purpose for human

presence there, science seems to have given way to tourism. In fact, from 1990, the number of tourists began to increase to a point where their numbers now exceed the number of scientists (Shaik, 2010). Initially, this was welcomed by research staff whose projects saw a potential finding boost through an increase in visitors, and scientists who stay in Antarctica all year round welcomed the idea of more human contact (Larson, 2012). However, it quickly became apparent that the large number of tourists was beginning to have negative impacts on the scientists. According to American University, which funds research in the Antarctica, independent tour companies are often unaware of research schedules and bring hundreds of tourists to scientific sites requesting explanations and tours (Grall, 1992). In addition, scientists staff sometimes have to stop their work to aid tourists who have either ignored the risks or underestimated the extreme conditions of Antarctica.

References

- Grall, J. (1992). Antarctic tourism impacts. *TED Case Studies*, 2(1). Retrieved from <http://www1.american.edu/TED/antarct.htm>
- Larson, S. (2012). More tourists head to Antarctica, affecting the region's ecosystem and science. *Peninsula Press*. Retrieved from: <http://peninsulapress.com/2012/05/11>
- Shaik, A. (2010). Antarctic wanderlust. *EJ Magazine*, 2010, Spring. Retrieved from <http://news.jrn.msu.edu/ejmagazine/2010/05/04/>

Unit 6 Making Oral and Poster Presentations

1

Critical Reading

Understanding the text

TASK 1

Purpose of the talk	to introduce nanotechnology
Key words	Nanotechnology; Principles; Applications
Definition of nanotechnology	Nanotechnology has the ability to reproduce things like diamonds and food to the world being devoured by self-replicating nanorobots.
Major benefits	It contributes to the development of medicine and protection of the environment.

TASK 2

- 1 A nanometer is even smaller than a hundred-thousandth the width of a human hair.
- 2 What we thought unbelievable can become a very real possibility.
- 3 They can replicate themselves and produce anything.
- 4 It can replicate diamonds, water and food. It can help the development of medicine and protect the environment.
- 5 There are enormous potentials and great interest in nanotechnology.

TASK 3

Paras.1-10

- 1○ 2○ 3○ 4 conduct 5○
6 drastically 7○ 8○ 9○ 10○ 11○

Paras.11-12

- 1○ 2○ 3○ 4○ 5○ 6○ 7 component 8○

Paras.13-15

- 1○ 2 suggest 3○ 4 hold up 5○ 6 scatter 7 extremely

TASK 4

- 1-5 3 1 5 2 4
6-10 9 8 6 10 7
11-14 14 11 13 12

TASK 5

- 1 The study of nanotechnology has attracted scientists from various disciplines, the joint research which has never happened before.
- 2 Many people believed that replicators could only exist in science fiction, but now it is possible to produce replicators.
- 3 Nanotechnology could render our dependence on non-renewable resources unnecessary.

TASK 7

Coordinated research is now being conducted in the burgeoning field of nanotechnology by scientists of biology, engineering and materials science. The presentation will first explain the concept of nanotechnology by detailing the key terms of nanometers and nanoscales. It then analyses the role of quantum mechanics and the way nanotechnology work. The main focus is the illustration of a vast range of the applications of nanotechnology. Specific examples are given in the field of cancer treatment and the environment.

Enhancing language ability

TASK 2

- 1 unprecedented, drastically, concept
- 2 caution, precise
- 3 exponential, realm
- 4 Additionally, diminished
- 5 foreseeable, eradicated
- 6 stick to, contradict
- 7 Conventional, dominates
- 8 programmed, automatically
- 9 scale, rearranged
- 10 exclusively, take advantage of, potential

TASK 3

- | | | | | |
|----------|-------------------------|-------------------------|-------------------------|--------------------------|
| 1 weight | 2 compose | 3 ☐ | 4 ☐ | 5 salary |
| 6 reduce | 7 ☐ | 8 ☐ | 9 design | 10 ☐ |

2 Academic Presentation

Doing research projects

- 1 The technology that assembles into others.
- 2 Because nanotechnology can be used to enable other kind of applications, e.g. daily products.
- 3 Sunscreen cream, golf clubs, clothing, battery etc.
- 4 Absorbing the UV light and keeping it from forming the damaging reactive oxygen species deeper into skin.
- 5 It introduces the nanomaterial-containing products.

Delivering an oral presentation

Choosing presentation styles

TASK 1

- 1 Unit 6 uses less formal and more colloquial style than Unit 4 and Unit 5.
- 2 The text of Unit 6 is less formal and more colloquial as the speaker wants to narrow the distance between the speaker and the audience.

- 3 Unit 6. Because the speaker faces the audience rather than readers, the audience who have no chance to raise any questions which can clarify what they fail to understand. Examples can help them gain a better understanding.
- 4 Unit 6. Because questions can engage the audience in the lecture.
- 5 Unit 6 uses the form of questions to outline the points the speaker may cover.
- 6 The colloquial style is used for two purposes: One is to engage the audience in the lecture; the other is to make sure that the audience understand the speaker.
- 7 Words and phrases like “OK”, “now let’s begin with”, “Now I’m going to” etc.

TASK 2

- 1 The study found out that this drug has serious side effects.
- 2 If we know its sources and symptoms, we can explain many things that otherwise we cannot analyze rationally.
- 3 If radiation was accidentally released, it will damage a wide area for a long time.
- 4 But the space skill can be applied in exploring our universe.
- 5 Crops grow quickly because the soil in this area is very rich.
- 6 Information has exploded and knowledge is increasingly obsolete. All this is forcing upon higher education a new set of demands.
- 7 The times signal that it is necessary to modify our legal system.
- 8 It is necessary to supervise the crops carefully so that they can survive when harvest time comes.

Making a good beginning

TASK 3

Infant-feeding practices in Shanghai; recent studies on infant-feeding practices; bottle-feeding and infant-feeding; first part / recent studies on infant-feeding practices.

Building a logical presentation in the body section

TASK 1

- 1 Three. They are: What is nanotechnology? How nanotechnology works? What are the benefits that we human beings can obtain from nanotechnology?
- 2 The speaker used such sentences like
OK, now let’s begin with the definition of nanotechnology.
OK, now I’m going to answer the second question: how does nanotechnology works?
Well, let’s move on the third question. How can we apply nanotechnology to benefit human beings?
- 3 Transitional words like *so, first, let me* and so on.

Using visual aids

TASK 2

Compared with the first one, the second slide uses a graph to present information. Thus there is a minimum effort the audience may make. It is unnecessary for the audience to read the slide. Instead, they could be listening to the presenter’s describing the slide. The second slide uses the graph which, more vivid than the table, helps the contents stand out.

Delivering your presentation

TASK 1

- 1 Yes. Because he wants to draw the audience's attention.
- 2 Yes. Because he wants to make his speech more dramatic and vivid.
- 3 He wants to emphasize the points he is talking about and provoke the audience's thought.
- 4 He wants to make emphasis on the points he is mentioning.
- 5 He wants to draw the audience' attention.

TASK 3

- | | | | | |
|-----|-----|-----|-----|------|
| 1 b | 2 a | 3 h | 4 f | 5 j |
| 6 d | 7 i | 8 e | 9 g | 10 c |

Appendix I

Sample Test 参考答案

1. 1-M 2-T 3-R 4-E 5-Q
 6-G 7-N 8-A 9-F 10-H
 11-J 12-B 13-O 14-S 15-I
 16-L 17-C 18-K 19-P 20-D

2.

- 1) Some people perceive radiation as a life force, others, however, believe that it will destroy the whole world.
- 2) According to other studies, some pre-service teachers believe that the depletion of the ozone layer is responsible for climate change.
- 3) The data-based presumption is that there might be two factors leading to the detrimental effect of GM soya on the newborn pups.
- 4) The necessity of establishing more rigorous systems is highlighted by the biostatisticians who are working for the organizations which have the systems for handling suspected fraud.
- 5) The wide application of nanotechnology in the technology market requires more efforts.

3. Literature review shows that there are quite different perceptions of GM foods in China although there is a tendency that more consumers will accept them.

4. Supplementary Foods Commonly Used by Study Sample

5. It seems that to reduce carbon emissions on a significant scale is rather difficult (McGuire, 2002). For one thing, although people in general are aware of the negative effects of global warming, they are uninformed of and even unconcerned with the cause of the climate change (Dalleva, 2003). For another, many countries such as the United States and Australia fail to make sincere and persistent efforts because of their “vested interest” (McGuire, 2002). The priority given to economy is also responsible for the difficult task (Biello, 2013). To maintain economic growth, people have to develop green energy, for example, “replacing coal-fired power plants with nuclear ones” (Biello, 2013).